

Linear combination

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$$v = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} ; \quad v_1 = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

$$\Rightarrow v = K_1 v_1 + K_2 v_2 \Rightarrow \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = K_1 \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + K_2 \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

$$-1 = 2K_1 - K_2 \quad \text{--- (1)}$$

$$2 = 4K_1 + 3K_2 \quad \text{--- (2)}$$

$$3 = -K_1 + 2K_2 \quad \text{--- (3)}$$

$$\begin{array}{l} \times 3 \rightarrow -3 = 6K_1 - 3K_2 \quad \text{--- (1)} \\ 2 = 4K_1 + 3K_2 \quad \text{--- (2)} \\ \hline -1 = 10K_1 \end{array}$$

$$\boxed{K_1 = -\frac{1}{10}}$$

$$\boxed{K_2 = \frac{4}{5}}$$

$$\begin{aligned} K_2 &= 2K_1 + 1 \\ &= 2\left(-\frac{1}{10}\right) + 1 \\ &= -\frac{1}{5} + 1 \\ &= \frac{4}{5} \end{aligned}$$

$$\begin{aligned} \text{(3)} &= \frac{1}{10} + \frac{8}{5} \\ &= \frac{1}{10} + \frac{16}{10} = \frac{17}{10} \neq 3 \end{aligned}$$

$$W = K_1 u + K_2 v$$

Example 2:

Consider the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$ in $\mathbb{R}^3$. Show that $\vec{w} = (9, 2, 7)$ is linear combination of $\vec{u}$ and $\vec{v}$.

Also prove that $\vec{w}' = (4, -1, 8)$ is not linear combination of $\vec{u}$ and $\vec{v}$.

$$\begin{cases} K_1 + 6K_2 = 9 \quad \text{--- (1)} \\ K_1 + 2K_2 = 1 \quad \text{--- (2)} \\ -K_1 + 2K_2 = 7 \quad \text{--- (3)} \end{cases}$$

$$\begin{aligned} &\xrightarrow{+} \begin{array}{l} 4K_2 = 8 \\ K_2 = 2 \end{array} \end{aligned}$$

$$\begin{aligned} -3 + 6(2) &= 9 \\ 9 &= 9 \end{aligned}$$

$$\begin{pmatrix} 9 \\ 2 \\ 7 \end{pmatrix} = K_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + K_2 \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}$$

$$9 = K_1 + 6K_2$$

$$2 = 2K_1 + 4K_2$$

$$7 = -K_1 + 2K_2$$

$$\boxed{W = -3u + 2v}$$

$$-K_1 + 4 = 7$$

$$-K_1 = 3$$

$$\boxed{K_1 = -3}$$

$$\begin{aligned} W' &= K_1 u + K_2 v \\ \begin{pmatrix} 4 \\ -1 \\ 8 \end{pmatrix} &= K_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + K_2 \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} \end{aligned}$$

Example 6: In each part the vectors are as linear combination of $P_1 = 2 + x + 4x^2$, $P_2 = 1 - x + 3x^2$ and $P_3 = 3 + 2x + 5x^2$

- a) $-9 - 7x - 15x^2$
- b) $6 + 11x + 6x^2$
- c) 0
- d) $7 + 8x + 9x^2$

$$6 + 11x + 6x^2 = K_1 P_1 + K_2 P_2 + K_3 P_3$$

$$= 2K_1 + K_1 x + 4K_1 x^2 + (K_2 - K_2 x + 3K_2 x^2) + (3K_3 + 2K_3 x + 5K_3 x^2)$$

Const

$$2K_1 + K_2 + 3K_3 = 6$$

$$x: K_1 - K_2 + 2K_3 = 11 \rightarrow \textcircled{1}$$

$$x^2: 4K_1 + 3K_2 + 5K_3 = 6$$

$$\begin{array}{ccc|c} + & - & + & \\ 2 & 1 & 3 & 6 \\ 1 & -1 & 2 & 11 \\ 4 & 3 & 5 & 6 \end{array}$$

$$= 2(-5-6) - 1(5-8) + 3(3+4) = -22 + 3 + 36 = -22 + 39 = 17$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 11 \\ 2 & 1 & 3 & 6 \\ 4 & 3 & 5 & 6 \end{array} \right)$$

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - 4R_1 \end{array} \begin{array}{ccc|c} K_1 & K_2 & K_3 & \\ \hline 1 & -1 & 2 & 11 \\ 0 & 3 & -1 & -16 \\ 0 & 7 & -3 & -38 \end{array}$$

$$K_1 - K_2 + 2K_3 = 11$$

$$K_1 + 5 + 2 = 11$$

$$K_1 = 4$$

$$3 \times \begin{cases} 3K_2 - K_3 = -16 \rightarrow \textcircled{1} \\ 7K_2 - 3K_3 = -38 \rightarrow \textcircled{2} \\ \hline -9K_2 - 3K_3 = -48 \\ -2K_2 = 10 \quad \boxed{K_2 = -5} \\ -15 - K_3 = -16 \quad \boxed{K_3 = 1} \\ -15 + 16 = K_3 \end{cases}$$